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3D Human Pose Estimation from multi-view thermal vision sensors

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ABSTRACT

Human Pose Estimation from images allows the recognition of key daily activity patterns in Smart Environments. Current State-of-the-art (SOTA) 3D pose estimators are built on visible spectrum images, which can lead to privacy concerns in Ambient Assisted Living solutions. Thermal Vision sensors are being deployed in these environments, as they preserve privacy and operate in low brightness conditions. Furthermore, multi-view setups provide the most accurate 3D pose estimation, as the occlusion problem is overcome by having images from different perspectives. Nevertheless, no solutions in the literature use thermal vision sensors following a multi-view scheme.

In this work, a multi-view setup consisting of low-cost devices is deployed in the Smart Home of the University of Almería. Thermal and visible images are paired using homography, and SOTA solutions such as YOLOv3 and Blazepose are used to annotate the bounding box and 2D pose in the thermal images. ThermalYOLO is built by fine-tuning YOLOv3 and outperforms YOLOv3 by 5% in bounding box recognition and by 1% in IoU value. Furthermore, InceptionResNetV2 is found as the most appropriate architecture for 2D pose estimation.

Finally, a 3D pose estimator was built comparing input approaches and convolutional architectures. Results show that the most appropriate architecture is having three single-channel thermal images processed by independent convolutional backbones (ResNet50 in this case). After these, the output is fused with the 2D poses. The resulting convolutional neural network shows excellent behaviour when having occlusions, outperforming single-view SOTA approaches in the visible spectrum.

1. Introduction

Thermal imaging has emerged as a game-changer in various fields, such as activity recognition, robotics, and surveillance [1–4]. Thermal vision sensors capture the heat radiation emitted by objects and accurately represent temperature and emission characteristics [4]. Among the main benefits of these compared to RGB, thermal images offer: (i) robustness to lighting conditions and clothing variations; (ii) the ability to capture the thermal signature of human body parts [5] and (iii) privacy preservation, as TIs does not reveal detailed physical features of individuals, making it a promising approach for privacy-preserving solutions, such as Ambient Assisted Living [6].

These benefits make thermal imaging a highly promising modality for 3D HPE [7]. The main goal of Three-Dimensional (3D) HPE is locating the 3D coordinates of a person's joints. It is currently being adopted in human motion analysis, user tracking methods, rehabilitation processes, or enhancement of surveillance systems.

Although robust 3D HPE solutions have been developed in the Visible Spectrum (VS), to our knowledge, there are no works in the literature using TIs. Most works using VSIs are Deep Learning (DL)-based solutions, using large annotated datasets to build robust and generalizable frameworks. Regardless, the big difference between VS and TIs features makes VS solutions unable to detect accurate 3D poses in TIs, and without annotated data, these are not easily transferable to thermal images [8].

Creating a 3D HPE dataset is a resource and time-consuming since it involves incorporating tags to the users' joints or manual annotation. In the literature, there is a lack of annotated 3D HPE datasets in the thermal domain. Therefore, developing a robust 3D HPE solution in the

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thermal spectrum is a challenging task that involves the creation of the dataset and the development of the pose estimation solution.

Regarding computer vision solutions applied to 3D HPE, key points can be extracted from one camera (single-view approach) or more than one camera (multi-view approach). Generally, multi-view solutions are more robust against occlusions and achieve more accurate poses since the user is recorded from different perspectives [9,10]. Although multi-view approaches achieve SOTA performance, these have not been applied to the thermal domain and will be handled in the current work.

In our work, we propose an end-to-end solution to perform 3D HPE in TIs. It includes an automatic approach to labelling TIs, removing human error and a costly setup. An IoT device incorporating visible and thermal cameras captures images, which are synthetically paired using the homography transformation. Thus, thermal images are labelled using visible SOTA algorithms. Furthermore, three devices have been deployed in the Smart Home of the University of Almería, building a multi-view scene to infer more accurate 3D poses. The proposed Convolutional Neural Network (CNN) uses the bounding box and 2D poses of the person from different perspectives in addition to the TIs to infer the 3D pose.

Summarizing, the main contributions of the current work are:

- Deployment of a multi-view setup with low-cost IoT devices (composed of Raspberry Pi, FLIR Lepton 3.5 and NoIR Camera) in the Smart Home of the University of Almería. Synchronization of the IoT devices using MQTT. The dataset is publicly available.
- Auto-labelling of the TIs, obtaining labels from visible-spectrum solutions such as YOLOv3 and BlazePose.
- Fine-tuning of YOLOv3 to crop the BB of the person in the thermal image (ThermalYOLO) and building a 2D pose estimator that obtains the 2D pose in thermal images to provide it as input to the 3D model.
- 3D HPE CNN, built from comprehensive experimentation including (i) input approaches and their permutations (3-channel image, 3x 1-channel images, and 2D key points), (ii) SOTA convolutional architectures (residual, inception, transformers) and (iii) learning approaches (from scratch, weights pre-trained and fine-tuning).

The remainder of the paper is organized as follows: In Section 2, we review related works and methodologies; in Section 3, we describe the devices, architecture, and methodology of the approach; in Section 5, we present the results; in Section 6, we detail our conclusions and ongoing work.

2. Related works

HPE is an active area of research that involves localizing the positions of human body joints in 3D space [11]. The technique has widespread applications in 3D movies, sports analysis, ambient assisted living, physiotherapy, and virtual reality [12–15]. In recent years, significant progress has been made in developing HPE solutions that provide detailed information about human body movements.

Before the advent of 3D HPE, researchers focused on developing 2D HPE models. These models can achieve good results and are relatively easy to train, as they do not account for occlusions between joints [16,17]. However, 2D models have limitations in complex real-world scenarios where occlusions are common. The advent of 3D HPE has enabled researchers to overcome this limitation and achieve better performance by accurately localizing joints in 3D space [18,19]. Recent developments in 3D HPE enhance the accuracy and effectiveness of HPE in various applications.

First, we introduce SOTA 2D HPE techniques, such as OpenPose [20], AlphaPose [21], and HRNet [22]. HRNet is a multi-resolution fusion approach that uses parallel streams of convolutional layers, each operating at a different resolution. The feature maps generated by each stream are then combined using residual connections to produce a final feature map that captures fine-grained details.

Newell et al. [23] introduced the Stacked Hourglass network, which employs convolutional layers and subsequent pooling and upsampling layers to predict the human pose. The input image is first encoded into low-resolution spaces, and then the features from these spaces are combined to obtain the final prediction.

Another approach to HPE is BlazePose [24]. This method uses a lightweight face-detector NN inspired by the Vitruvian Man to detect the person in the input image. Then, another NN identifies the 3D landmarks of the person.

OpenPose [25] is another widely used solution that utilizes a fine-tuned VGG-19 NN [26] to obtain the part affinity fields in the input image. A greedy algorithm is then employed to link the different body joints of the people.

The aforementioned solutions have been trained to handle VSIs. However, their performance on TIs is far from being well. The reason for this is the different domain, and the fact that the datasets used to train them do not contain TIs. To fill this gap, authors in [27] proposed THPoseLite. In this work, authors evaluate BlazePose [24], and OpenPose [25], and no more than 50% of the poses were obtained on TIs. Therefore, they developed a 2D HPE solution on TIs. The solution involved a pre-processing of the TI, removing the background using a motion estimator. This preprocessing allowed the VS pose estimator (BlazePose) to extract the 2D pose in more than 75% of the TIs. Then, having the 2D poses from BlazePose, a regression neural network was built, using MobileNetV2 as the backbone.

Despite recent advancements in 3D HPE, it remains challenging due to the requirement of specialized sensors, such as depth sensors and IMUs, for capturing data in controlled environments [28,29]. While these solutions can achieve high accuracy in such settings, they often struggle to generalize to non-controlled environments.

Regarding categorization, 3D HPE solutions are grouped based on the number of cameras used for capturing images and the number of individuals the models can recognize. Single-view solutions, such as MonoPerfCap [30,31], use only one monocular Red, Green and Blue (RGB) camera and can recognize a single person (single-view single-person) or multiple people (single-view multi-person). In contrast, multi-view solutions, such as [32,33], employ multiple cameras and can detect single or multiple individuals. These solutions are designed to handle occlusions and can achieve better accuracy in complex scenes.

Single-view methods for 3D HPE are limited in handling occlusions. When using only one camera, it is difficult to disambiguate between different possible 3D poses that correspond to the same 2D projection. Existing solutions can be classified into direct estimation methods and 2D to 3D lifting methods [34].

Direct estimation methods for 3D HPE infer the 3D pose directly from the input image without first estimating the 2D pose. These methods involve training NNs to minimize the error of generated 3D joints by taking an RGB image as input with 3D joints as ground truth. One example is the network proposed in [35], which consists of a convolutional part followed by two subnetworks that transform image features into joints. Other methods, such as the one proposed in [28], discretize the 3D space around the subject and train a CNN to predict per voxel likelihoods for each joint. This approach allows for a more natural representation of the 3D pose and results in better performance. In [24], the authors propose BlazePose, a lightweight face-detector NN to detect a person and then identify landmarks with another network to estimate the 3D pose of a person's body. With real-time processing capabilities, BlazePose achieves SOTA performance in 3D HPE. In [36], the authors propose a method that leverages RGB images to estimate the 3D human pose directly training a CNN to generate 3D joint coordinates from the input image.

The 2D to 3D lifting approach is another solution for estimating 3D human pose. These solutions begin with a 2D pose, which is then transformed into a 3D pose using SOTA 2D HPE methods as a starting point. Because the 2D baseline is highly accurate, it is easier to construct a

3D pose than to start from scratch. This approach typically outperforms direct estimation methods, and several advanced 2D HPE methods such as [22,23,25] have been widely used as 2D pose detectors in 3D HPE methods. For example, [37] utilizes 2D key points obtained from [23] to generate multiple feasible hypotheses of 3D pose using multimodal mixture density networks. Another solution, presented in [38], uses 2D key points obtained from [22] to infer a 3D pose from motion patterns observed in a video sequence.

Single-view thermal imaging has been gaining popularity for 3D HPE due to its ability to capture human body temperature and provide unique features complementary to RGB images. In [39], the authors propose a 2D to 3D lifting method that utilizes thermal and RGB images to improve the accuracy of 3D HPE. Specifically, they use an RGB image to estimate the 2D pose and then lift the 2D pose to a 3D pose using the TI as an additional input. This approach achieves SOTA performance on the HumanEva-I benchmark dataset. Similarly, in [40], the authors use a graph-based local-to-global network to process 2D human pose sequences and estimate the 3D pose represented by the 3D coordinates of 17 joints.

The problem of partial occlusion presents a significant challenge for single-view 3D HPE. To address this issue, a natural solution is to estimate the 3D human pose from multiple views. This is because the occluded parts in one view may become visible in other views, which allows for the reconstruction of the complete 3D pose. The challenge, however, is to accurately associate the corresponding location between different cameras to achieve an accurate 3D pose reconstruction.

Several methods have been proposed to tackle the association problem. One group of methods used body models to optimize model parameters to match the model projection with the 2D pose [41–45]. Although these methods are effective, they require large memory and have an expensive computational cost.

Alternatively, multi-view matching frameworks [46–49] aim to reconstruct the 3D human pose across multiple viewpoints using consistency constraints. While these frameworks aggregate 2D keypoint heatmaps of multi-view images into a 3D pictorial structure model, they require the model to be retrained when the multi-view camera environment changes. In contrast, multi-view geometry combines information from multiple views to improve the accuracy of estimated poses, especially in crowded scenarios where single-view methods may fail due to occlusion and ambiguity. Chen et al. [50] propose a maximum a posteriori method to improve the triangulation process for multi-person 3D HPE in crowded scenes. Qiu et al. [51] propose a cross-view fusion method that achieves SOTA performance on various benchmark datasets. Zhou et al. [52] introduce a novel voxel-based method for multi-camera 3D HPE in wild environments, outperforming SOTA approaches on several datasets. Spurlock et al. [53] propose a multi-modal 3D HPE framework that can effectively fuse information from multiple sources, including RGB images, depth maps, and temporal information, achieving superior performance on various datasets. Kocabas et al. [54] introduce a self-supervised method for 3D HPE that leverages multiple-view geometry to achieve SOTA performance on several benchmark datasets. Dong et al. [55] presents a multi-way matching algorithm to cluster detected 2D poses across multiple calibrated camera views, enabling effective inference of 3D poses. Zhang et al. [32] directly regresses the 3D joint locations without intermediate tasks, using learnable query embeddings and projective attention. It achieves SOTA performance on benchmarks, including a 9.8% improvement on the challenging Panoptic dataset. These approaches represent promising insights for multi-view 3D HPE and provide ideas for future research, going a step further with the inclusion of thermal imaging.

Comparing the results from the scientific literature [7,11,16,34,56–58], it is evident that the performance of multi-view 3D HPE methods has consistently shown improvements compared to single-view 3D HPE methods using the same dataset and evaluation metric. In addition, the multi-view setting has been shown to help alleviate issues such as occlusion and depth ambiguity. These works have reported on the SOTA in multi-view 3D HPE, highlighting the benefits and challenges of this approach and recent advances in methods and techniques. Overall, these studies provide strong evidence for the effectiveness of multi-view 3D HPE methods and suggest that they will continue to play a critical role in advancing the field of HPE.

Despite the high performance achieved by DL-based solutions in HPE, challenges such as insufficient training data, depth ambiguities, and occlusion still exist. Additionally, it is worth noting that many of the methods proposed in the literature have limited practical use due to the complexity of their implementation. In many cases, the datasets and adaptations made to the models are not publicly available, making it difficult for other researchers to reproduce the results. While private datasets may result in better performance, public datasets allow for reproducibility and comparison with other methods. Even when the datasets are available, they are often over-trained and unable to generalize to other types of images, requiring manual adjustments of camera calibration parameters to achieve acceptable results. These issues have been identified in previous studies, for example, [15,28,59], highlighting the need for more accessible and generalizable solutions in HPE field.

3. Methodology

In this Section, the proposed framework is thoroughly explained. First, the camera setup where the dataset was recorded is described. Second is the overall architecture proposal with all the steps to build the solution. Finally, the different phases shown in the architecture proposal are explained in depth.

3.1. Cameras setup

This work takes images from different perspectives in the visible and Thermal spectrum (TS). The IoT device that was utilized to capture thermal and VS images from a point is made up of two cameras: (1) a 160×120 pixel thermal vision camera, such as the FLIR Lepton 3.5 with the Pure Thermal 2 board, and (2) a VS camera, such as the Raspberry Pi NoIR Camera Module, with a resolution of 2592×1944 pixels. Both cameras have been embedded in a 3D printing case attached to a Raspberry Pi 4 device to make it easier to match the images regarding height, placement, etc. In addition to the cameras, a fan has been attached. Fig. 3 shows the IoT device and the scheme of its components.

Three IoT devices have been built and deployed to record images from three different perspectives in the Smart Home of the University of Almería. These devices have been fixed to a height of 1.90 m. The initial distribution of them was to build an equilateral triangle between them. Nevertheless, it was impossible to situate them in that way. Their final distribution can be appreciated in Figs. 1 and 2.

In this work, the 3D pose is reconstructed using images from three different perspectives. Therefore, cameras have to be synchronized to record images at the same timestamp. To achieve this, the MQTT protocol was used. Every 0.3 s, the Smart Home's server sends the current timestamp to the three IoT devices. The cameras record both images as soon as the MQTT message is received.

When the 3D pose recognition model is deployed and running at the Smart Home's server, the IoT devices send the recorded TIs using MQTT to the server. These images are aggregated in the server and evaluated using the model.

In this way, the server and the three IoT devices establish a bidirectional connection using MQTT. It is important to remark that all communication remains in the local environment without external services.

3.2. Solution architecture proposal

In the proposed approach, unpaired visible and thermal images are processed in several phases whose main goal is to train a NN

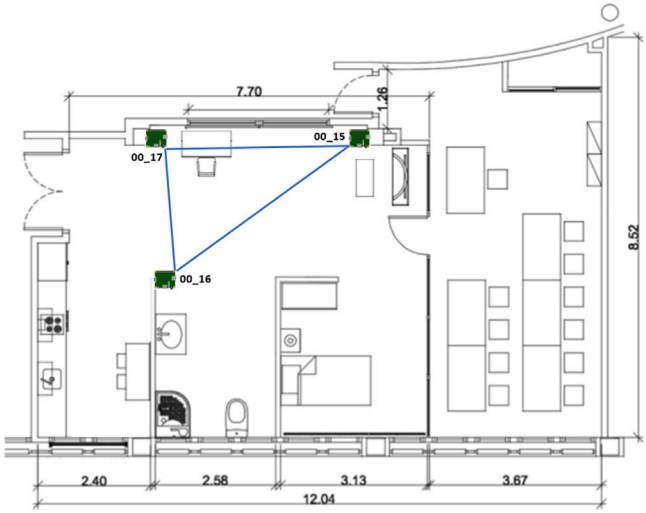

Fig. 1. Distribution of the IoT devices.

Fig. 2. Location of the IoT devices.

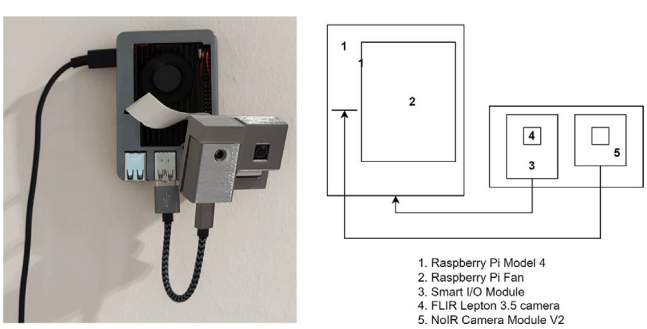

Fig. 3. Components of the IoT device.

model that extracts 3D poses from thermal images. These phases can be depicted in Fig. 4.

The homography matrix is calculated from an unpaired dataset to map the VSIs to the TIs. Then, state-of-the-art visible spectrum solutions are used to extract the bounding boxes (YOLOv3), 2D and 3D poses (BlazePose) on VSIs. As the methods used in the visible spectrum solution do not have a high performance on thermal images, some customized solutions have been proposed to work using only TIs as input (instead of VSIs). The VSIs and annotations of these using VS solutions are only used when the TS solutions are being trained. First, ThermalYOLO is the method charged to obtain the BB from the TIs. It is a YOLOv3-inspired NN and uses TIs and the BB extracted from YOLO on the paired VSIs in the training phase. Thus, in inference mode, bounding boxes are obtained using only TIs. Then, these images are pre-processed, cropping the user and centring it on the image. After this, a 2D pose estimator is trained using the TIs and the 2D poses obtained in the visible spectrum. 2D pose estimator receives a TI without background and containing a person centred and obtains the 2D pose. Finally, from the TI and 2D key points, a 3D pose estimation NN is built, generating 3D poses. Following the same methodology as in the previous case, the 3D pose estimator is trained using TIs and the 3D poses obtained by BlazePose on the paired visible spectrum images.

3.3. Phases

In this section, the different phases of the proposed solution are thoroughly described. First, the homography calculation; second, the fine-tuning of YOLO; third, the 2D pose estimator and finally, the 3D pose estimator.

3.3.1. Pairing visible and thermal cameras

The intrinsic and extrinsic parameters of the cameras are inherently different. Thus, although cameras are physically aligned, images obtained by the cameras are not entirely paired.

To obtain fully paired images, the use of the homography [60] is proposed. Homography enables the mapping of points from a source image plane to the target image's matching plane. This method is frequently used to find a certain object in an image. Thus, the 3×3 homography matrix, given by Eq. (1), is used to transform the points of the source image into points in the target image. Point (x_s, y_s) is a point in the source image, and point (x_d, y_d) is a point in the target image. The matrix H maps these points.

$$\begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix} = H \begin{bmatrix} x_d \\ y_d \\ 1 \end{bmatrix} = \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_d \\ y_d \\ 1 \end{bmatrix} \tag{1}$$

The values of the H matrix are calculated by the least squares method, using a minimum of 4 points in both images [61]. An alternative is to use the SIFT key point search [62] and then find the homography matrix with the RANSAC method [63]. Another alternative is the 2D HPE between the visible and TIs. However, these approaches performed worse in preliminary experiments, mainly since, nowadays, key point search in the thermal imagery domain is not widely used.

Fig. 5 shows the transformation of the RGB image to have the same perspective as the thermal one. It is important to note that points from the person and other objects, such as a computer, table or fridge, have been manually annotated. Otherwise, if the points refer only to the person, the resulting image would fit perfectly with the person in the TI, but the remaining objects do not.

3.3.2. ThermalYOLO

The first step when detecting the 2D and 3D pose in several solutions is obtaining the BB of the person in the image. This is not only to help the pose detector to locate the person in the image and avoid problems with the different locations of users in the image. Using the BB to extract the person in the image allows the current approach to be executed in multi-person scenarios. In the thermal domain, no solution has been specifically designed to achieve it. Despite not being trained using TIs, SOTA VS solutions such as YOLO can obtain the BB in some of them. However, its performance in TIs is far from that of VSIs.

In this work, a new solution, ThermalYOLO, is proposed to fill the gap between the thermal and visible domains in terms of people BB detection. YOLO is used to extract the BB of people in the transformed VSIs. As the VSIs and TIs are paired using the homography, the BBs

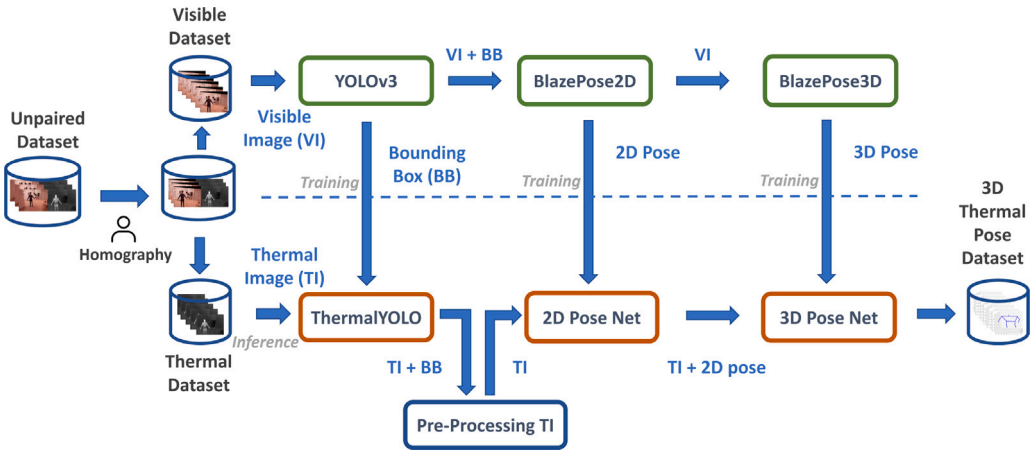

Fig. 4. System architecture.

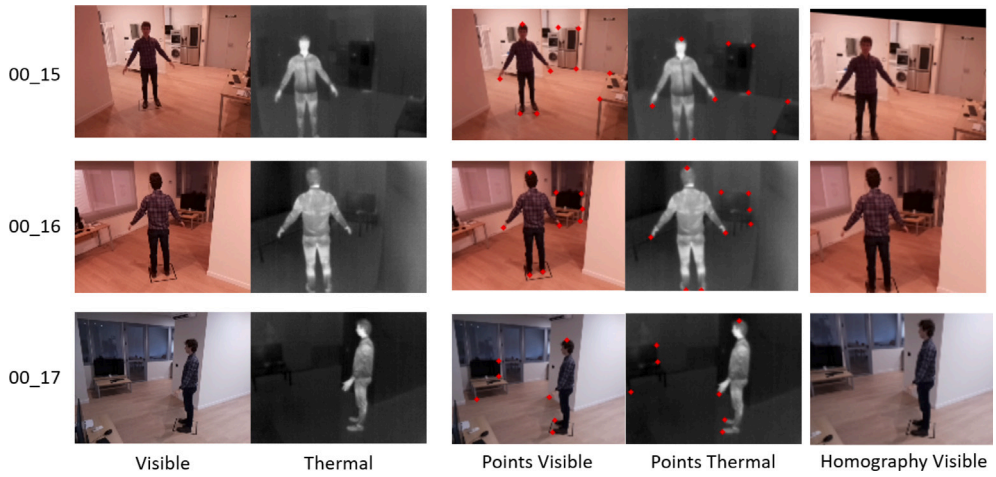

Fig. 5. Perspective transformation using homography.

from the VSIs can be used to annotate the TIs. Thus, a fine-tuning of YOLO is carried out using the annotated TIs. The new fine-tuned YOLO NN is called ThermalYOLO. When ThermalYOLO is trained, in a real scenario, ThermalYOLO is able to detect the BB of people in TIs, avoiding the use of VSIs.

Nowadays, there exist several object detection approaches [64]. First, the two-stage detectors try to locate some object proposals in a first inference, and then, classify and locate these proposals in order to detect the objects present in the image. These methods are heavier to train and execute, but used to achieve SOTA performance. Mask R-CNN [65] and Faster R-CNN [66] stand out in this category. Second, single-stage detectors locate and classify objects using a single step using different grids and pre-defined boxes. These solutions are more lightweight, achieving SOTA performance in terms of real-time object detection, but their accuracy is lower than the two-stage methods. The best-performing single-stage detectors are YOLO-like methods such as YOLOv4 [67] and YOLOv7 [68]. Third, transformers have also been used in this task due to their general-purpose nature. Currently, solutions such as Co-DETR [69] achieve SOTA performance,[1] although they are composed of a large number of parameters.

Among all the solutions, YOLOv3 [67] achieves good performance and the inference time is negligible. In addition, it is available publicly, and it is easy to integrate into Deep Learning projects since the pre-trained model is available in TensorFlow and Python.

YOLOv3 divides the image into a grid of SxS dimensions, where each cell is responsible for predicting the BB of objects whose centre falls in that cell. The maximum number of objects is a user-defined parameter. It defines the complexity of the NN since the greater the number of objects, the better it will be able to detect small objects close together, but it will increase the size of the network and the training time. Thus, in addition to obtaining the bounding boxes with a centre, YOLO provides a confidence value for each BB indicating the certainty that the network has about the predicted BB. The value of confidence is between 0 and 1, being 1 the maximum confidence. The NN architecture contains 24 convolutional layers and two fully connected layers. The NN has been trained on the COCO dataset, predicting up to 80 objects.

The output of the YOLO NN consists of a set of bounding boxes and the corresponding confidence values. The NN typically generates more BB estimates than actual items in the image. Typically, a real object has more than one BB. Consequently, it is required to do a post-processing stage that can look through the bounding boxes the NN generated to find those that match the actual objects in the image.

One of the most commonly used techniques is the Non-Maximum suppression [70]. This technique reviews the list of bounding boxes ordered from highest to lowest confidence value. The Intersection Over Union (IoU) value concerning the previous bounding boxes is computed for each BB. The IoU value (Eq. (2)) is the quotient between the area formed by the intersection of two sets and the area formed by the union of these sets, which is related to the Jaccard index. If this value is lower than a threshold set by the user, the BB is different from the

[1] https://paperswithcode.com/sota/object-detection-on-coco.

rest and can be considered a new object. Otherwise, the BB is inferred that corresponds to the same object. In this case, after some preliminary experimentation, it has been considered to set the threshold IoU value to 0.4.

$$IoU(A, B) = \frac{A \cap B}{A \cup B} \tag{2}$$

ThermalYOLO is trained using all the TIs whose paired VSI contains a person after being evaluated by YOLO. As for the training procedure, several approaches are compared:

- **From Scratch**: The initial weights are random. All the layers are trained.
- **Convolutional weights pre-trained**: The convolutional weights are pre-trained on the COCO dataset and frozen during the training. The fully connected upper layers were not trained.
- **Fine-Tuning**: Similar to the previous approach, however, the values of the fully connected upper layers were also pre-trained on COCO. The convolutional layers are also frozen.

As a result, ThermalYOLO is adapted to the TI domain and can infer the BB of people in TIs. Furthermore, obtaining the BB of the different people in the image enables the top-down approach in HPE. This means that the proposed solution can obtain the pose of multiple users in the same scene, as each is processed independently.

3.3.3. 2D pose estimator

Once the bounding boxes are obtained in the single TI from the three different perspectives, the 2D pose of the user situated in the BB is inferred using a 2D HPE NN. The 2D pose from different perspectives is used in the proposed 3D HPE NN to infer the 3D pose. Only the most relevant key points and joints are selected in this case. The key points follow the COCO format.

Similar to the BB estimation, visible spectrum 2D HPE frameworks do not perform well on TIs, as stated in [27]. Therefore, developing a 2D HPE solution is necessary in this work.

Different from [27] that preprocess the TI to increase the 2D pose detection using BlazePose on TIs, in this case, key points are extracted directly using BlazePose on the paired VSIs. Thus, more correct poses are obtained (since BlazePose was trained on VSIs), allowing the training of a more robust solution.

In [27], a motion estimation algorithm was used to remove the background of the TI. This preprocessing improved the performance of the 2D pose estimation algorithms trained on VSIs. In the current work, this is not necessary because the ground truth 2D pose is obtained in the paired VSIs. However, TIs are preprocessed in order to increase the accuracy of the proposed 2D HPE NN on the input TIs. The preprocessing (Fig. 6) involves the following steps:

- Crop the BB: ThermalYOLO obtains a BB containing the person in the image. This BB is cropped, and the rest of the image is removed.
- Centre the crop in the image: The BB is embedded in the centre of a new image of size 120 × 120. In addition, at least 10 pixels of padding are added to all the image edges. The background is set as black.

The training steps that are required to train the proposed 2D HPE NN are the following:

- Obtain the 2D poses in the VSIs transformed with the homography using BlazePose.
- Pre-process the TIs following the steps mentioned before. The resulting image is the input of the 2D HPE NN.
- Map the pose key points from the VSI in the new centred TI. This is the ground truth or label of the input-centred TI.

Therefore, the NN takes the centred TI as input and predicts a 2D pose. We propose a convolutional-regression-based NN among all the possible NN architectures. The first layers are convolutional, aiming to extract the main spatial features. After these, some fully connected layers are concatenated, ending with a layer containing 34 neurons. These neurons map the key points in the following way: $[x_0, y_0, x_1, y_1, \ldots, x_n, y_n]$, being n the total number of key points to predict.

The architecture of the 2D pose estimator model is shown in Fig. 7. At the top, the convolutional part is incorporated, extracting spatial features of the input image and producing a 26 neurons output. After this, three fully connected layers of 512, 128, and 64 neurons are incorporated. These have ReLU as an activation function.

3.3.4. 3D pose estimator

In this work, the main goal is to extract the 3D pose of a person in a scene using several thermal cameras. None of the tested 3D HPE NNs available in the literature achieve it. Therefore, we propose a new 3D HPE NN fusing images from different perspectives and regressing the 3D key points of a person.

First, to train the proposed NN, the ground truth 3D key points are extracted using available solutions in the literature. These solutions only work in VSIs, which are applied to the paired VSIs transformed with the homography.

More specifically, the 3D key points using the three cameras are obtained following the format $[x_0, y_0, z_0, x_1, y_1, z_1, \ldots, x_n, y_n, z_n]$, being n the total number of key points to predict. On the one hand, VoxelPose uses the three cameras to infer the 3D key points. On the other hand, BlazePose obtains the 3D key points from a single image, so 3 sets of 3D key points are obtained in the proposed camera setup. In this case, the ground truth comprises the key points whose camera obtains more poses. It is important to remark that the difference between the 3D points obtained by BlazePose and VoxelPose is developed in Section 5, and only key points from one solution are taken as ground truth.

In this work, the 3D HPE network can contain several types of inputs: (i) a 3-channel image consisting of the 3 single-channel TIs from the three thermal cameras of the setup, (ii) three independent single-channel images, and (iii) 3 sets of 2D key points. TIs have the same pre-processing as the 2D pose estimator described before.

To better assess the importance of each input and its permutations, the following configurations have been evaluated:

(A1) **3-channel image**: The frames from the three cameras are fused in a single input. The convolutional filters are common to the three TIs. These filters extract the spatial features of the input image. The convolutional part comprises a well-known architecture like those presented in previous sections. After the convolutional part, some fully connected layers are concatenated to make the regression of the 3D keypoints.

(A2) **3-channel image + 2D key points**: The same configuration as before but also incorporates the 2D key points as input. The 2D poses are linked to the convolutional spatial features before the fully combined layers. These key points would help to better locate the 3D key points in the image, as the 2D coordinates of one camera would be similar to the x and y coordinates of the 3D pose, and the other 2D key points would help to infer the z coordinate in the 3D pose.

(B) **2D key points**: In this configuration, TIs are not used to infer the 3D pose of the person, only the 2D key points. Although no images are incorporated as input, the key points are processed with convolutional layers. Architectures integrating 1D convolutional layers, 2D convolutional layers, and residual blocks are compared. After the convolutional layers, the regression fully connected layers are concatenated.

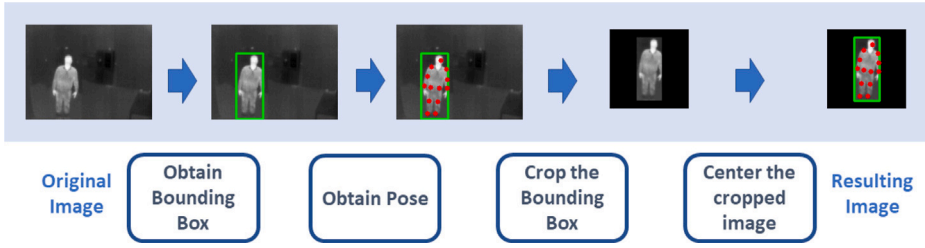

Fig. 6. Pre-processing of the TIs.

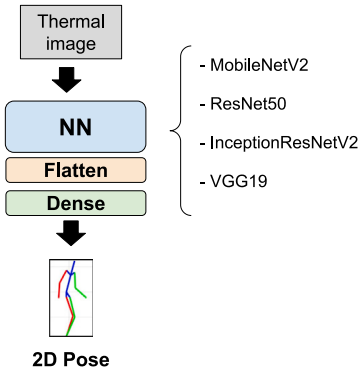

Fig. 7. 2D HPE NN architecture.

(C1) **3x 1-channel images**: Each TI is processed independently using three separate CNNs, so their filters are not typical and can extract specific features of each camera. In this case, the number of filters in each convolutional network is divided by three concerning the configuration using a single 3-channel image. It is important to remark that the convolutional networks are the same. The result of the pre-processing of each image is concatenated into a single vector at the end. It does not make use of the 2D pose.

(C2) **3x 1-channel images + 2D key points**: The same configuration as above, but incorporating the 2D key points to the feature vector containing the spatial information of the input images.

Fig. 8 shows a scheme of the different 3D NN configurations with the independence of the resulting specific convolutional architecture selected.

Concerning the specific CNN used in this work as backbone architectures, we develop and evaluate several pre-trained and from-scratch NN architectures of varying complexity (Fig. 9).

- **Pre-trained models**: These pre-trained models are available in the Keras[2] library after being trained on ImageNet. In our case, only the convolutional part is extracted to build our custom solution.

  (A) **ResNetV50** [71]: Residual NNs allow building complex and deep NNs by incorporating identity blocks with skip connections that palliate the gradient vanishing. Convd1x1 and Conv3x3 are general spatial extractors with batch normalization and relu. A reduction block with stride is applied to reduce the input size along the network. Between the wide range of residual networks, ResNetV50 offers an encouraging configuration that balances performance and complexity.

  (B) **MobileNetV2** [72]: MobileNetV2 is a NN specifically designed to achieve SOTA image segmentation and object detection when executed on mobile devices. It incorporates point-wise and depth-wise convolutions that enable light filtering, reducing the cost of the convolution. Its good performance in image segmentation would allow this architecture to deal effectively with the HPE problem, as the main convolutional layers are required to extract spatial features from which the 2D Pose key points are extracted. Similarly to ResNetV50, it is made by blocks (residuals from v2) interleaved by reduction blocks with stride.

  (C) **InceptionResNetV2**: This NN includes inception and residual blocks in the design. Three blocks, IR1, IR2 and IR3 are configured to extract spatial features from low-level to high-level layers. In addition, reduction blocks with stride are interleaved in the deep design by reducing dimension size.

- **From-scratch models**: These models are not pre-trained nor available in the Keras library. Therefore, transfer learning cannot be applied to them. In addition, these contain convolutional blocks and fully connected layers that regress the spatial features extracted from the convolutional blocks.

  (D) **Vision transformer model** [73]: A transformer model is evaluated to the 3D HPE from the TIs of multi-camera devices. This challenger goal without 2D landmarks using the raw thermal data as a transformer vision model faces input [73]. In this model, four transformer blocks with modified self-attention computation (Swin Transformer blocks) are connected to extract spatial features. Self-attention in non-overlapped windows and shifted window partitioning in the successive blocks are implemented regarding the approach [73].

  (E) **10-layers Conv1D model (TermalPose3D-2D-conv1D)**. This lightweight NN models the 2D landmarks from multi-cameras TIs to compute the 3D HPE. It includes five double layers using Conv1D from input channels comprising 2D landmarks for each camera.

  (F) **InceptionResnet Conv1D model**: This inception NN incorporates filters with multiple sizes executed on the same level. These obtain wider rather than deeper NNs. These allow capturing different size features in Conv-1D from 2D landmarks for each camera. Furthermore, this version incorporates some residual skip connections, interconnecting low-level and high-level features. Two blocks (R1 and R2 with input size reduction) are interleaved in the shape R1-R2-R1 connection. Average pooling and dense layers are connected to model the 3D HPE.

4. Experimental setup

This section describes the dataset used to validate the proposed methodology. Then, the metrics adopted to compare the different solutions are also defined.

[2] https://keras.io/api/applications/.

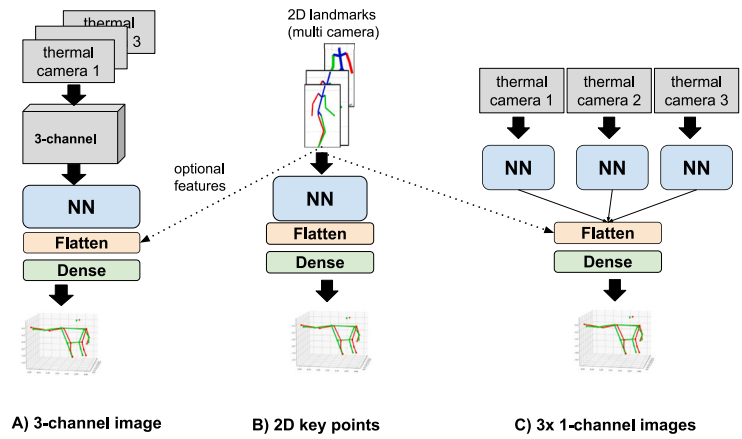

Fig. 8. 3D HPE NN configurations.

Table 1
Dataset description.

Devices used	3x IoT devices
Image dimensions	160 × 120
Number of images	18,454 × 3 cameras
Number of people	16 (13 males, 3 females)
Conditions	Temperature: 19–21 °C Morning

4.1. Dataset

To validate the proposed approach, relevant literature was reviewed to obtain TI datasets of people from different perspectives. Despite this search, no suitable datasets or works were found. For this reason, creating a custom dataset to facilitate training and evaluation of the proposed approach was required. The key characteristics of the resulting dataset are presented in Table 1.

The dataset was gathered in the Smart Home of the University of Almería, and it features 16 participants, including 3 women and 13 men, ranging in height from 1.55 to 1.93 m. In this case, 75% of the participants were undergraduate students. Participants were instructed to perform a kind of choreography, raising arms and legs, making some strides, bending, hiding the face with the hands, and moving in the scene. This variety of movements allows us to have movements involving most of the axes and kinds of poses that can be obtained when the user is interacting with other people or with appliances in a controlled environment. Cameras recorded paired images at a frequency of 3 FPS. The total recording time was around 1.7 h. To better validate the proposed approach, the leave-one-subject-out approach was carried out. In this case, 14 users were incorporated in the train and validation dataset, while 2 users were set as the test dataset. Therefore, when building the proposed method, three datasets were created: (i) training dataset, which has 80% of the images from 14 users; (ii) validation dataset, containing 20% of the images from 14 users and (iii) containing 100% of images from 2 different users. This evaluation methodology enables the training and validation using data from 14 users, but keeping two users isolated to test every model with new unseen data different from the training and validation.

The dataset is publicly available in Mendeley Data [74].

4.2. Metrics

This work involves the comparison between different person detection solutions, 2D and 3D pose estimation methods. Each of the problem can be measured with different metrics.

First, person estimation solutions make use of two metrics to compare the performance: the percentage of people and the IoU value. The percentage of people indicates the percentage of images in which the proposed solution produces an object. This metric is useful because the first step in the 3D pose estimation algorithm is cropping the bounding box of the person. A solution that is not able to detect the person is not adequate for this problem. The IoU value (Eq. (2)) indicates the quotient between the area formed by the detected person and the ground truth bounding box. When a person is detected, this metric measures the precision of the recognized person. A higher value indicates a better intersection between the observed person bounding box and the bounding box indicated by the model.

Second, pose estimation solutions are compared using the Root Mean Squared Error (RMSE). Generally, this value measures the average deviation between the predicted and actual values in a dataset. Thus, it can be used to assess how closely the estimated pose matches the ground truth pose. The equation is defined in Eq. (3). In this case, we suppose that each pose produces one RMSE value. N defines the total number of keypoints multiplied by 2 (coordinates x and y) that each pose has; y_i indicates the real value and $\hat{y}_i$ the predicted value. When the solutions are compared in the following sections, the RMSE shown is the average between the RMSE values of the different images of the dataset.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{3}$$

Similar to RMSE, the RMSE per pixel is used to compare RMSE values between images having different resolutions. This is useful because RMSE values from different resolutions cannot be fairly compared. The same RMSE value in two images having different resolutions, indicates that the model using images containing the images with higher resolution is better, as it is more difficult to place key points in bigger images and a visually negligible difference can result in a great number of pixels.

All the metrics defined above are used to compare the performance and suitability of the different solutions. Besides, other metrics are used to measure certain aspects of the deep learning models used on them. The number of parameters (NP) defines the total number of parameters of the deep learning model; the time per epoch (TE) is the time required to train the model during one epoch; the number of epochs (NE) is the total epochs that are used to train the model; TL and pre-trained indicates if the model was trained using transfer learning (yes) or from scratch (no).

5. Results

In this work, the main goal is to estimate the 3D key points of a person using three thermal cameras. To achieve this, the proposed method

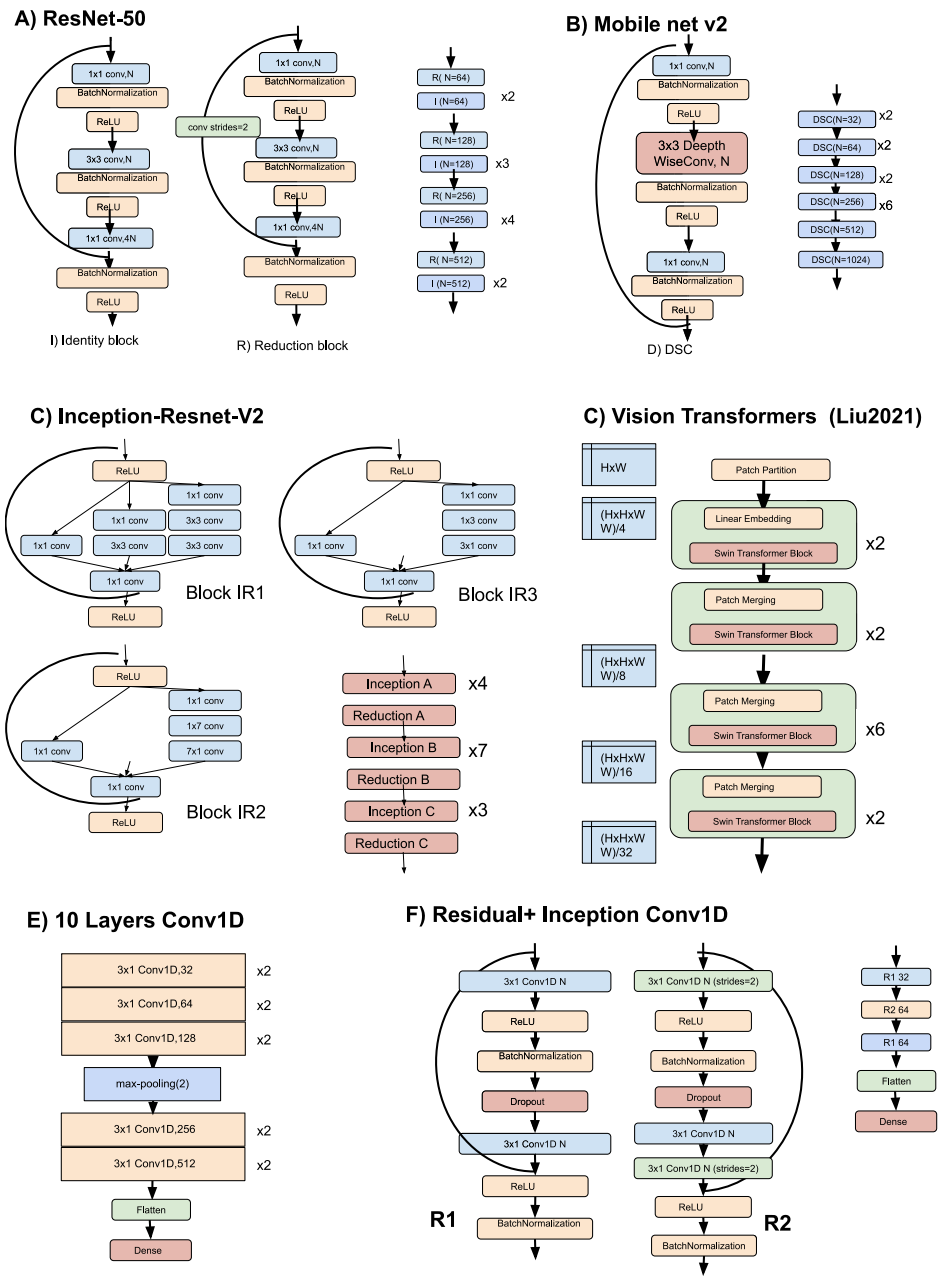

Fig. 9. CNN models: (A) ResNet-50; (B) Mobile net v2; (C) Vision Transformers; (D) Inception-Resnet-V2; (E) 10 layers Conv1D; (F) Residual + Inception Conv1D.

incorporates several phases such as (i) obtaining the BB, (ii) extracting the 2D pose, and finally, (iii) estimating the 3D pose. This section includes the experimentation developed to support our approach.

5.1. Obtaining BB

First, following the general scheme in Fig. 4, the person is cropped from the TI using the YOLO NN. Since the people detection of YOLO on raw TIs is far to be similar to the performance on VSIs, the training of YOLO on TIs (ThermalYOLO) was proposed.

When training ThermalYOLO, three different configurations were compared. Table 2 shows their performance in terms of the number of epochs, time per epoch, and training/validation error. The configuration 'Scratch' achieved the lowest values in training and validation error, being the most appropriate configuration by far. Although the error is higher in the case of 'conv pre-trained' and 'fine-tuning' (close to 7 in training and 7.5 in validation), they require less time to train (less than 15 min per epoch). The reason is that these configurations have the convolutional weights frozen, avoiding backpropagating errors in the convolutional part, different from the configuration from scratch.

More deeply, the 'conv pre-trained' and 'fine-tuning' configurations are supposed to perform better than the 'Scratch' configuration in most computer vision problems. In these cases, the convolutional layers are trained on ImageNet, containing millions of VSIs. Thus, these blocks have learnt common patterns and features from images in the same domain as Imagenet. This knowledge can be used to train specific solutions without learning similar patterns again. Nevertheless, in this case, TIs have a completely different domain, so the distribution learnt by pre-trained NNs is very different to the distribution of the TIs. Thus, pre-trained NNs struggle to learn a completely different distribution, so it is better to train from scratch. Thus, it can be appreciated that TIs have particularities that make DL solutions on VSIs not perform well on different domains, being necessary to enrich the training data of these, to make them more robust in different domains.

Fig. 10. ThermalYOLO's performance on test images.

Table 2
Comparison of ThermalYOLO training approaches.

Name	NE	TE	Train error	Val. error
Scratch	46	15 m 30 s	**0.3818**	**0.5735**
Conv pre-trained	47	**13 m 14 s**	7.1351	7.5655
Fine-tuning	**24**	14 m 37 s	7.0874	7.7498

Table 3
YOLO and ThermalYOLO's performance.

	YOLO version	Train/Val		Test	
		Percentage	IoU	Percentage	IoU
Visible	YOLO	0.99869	–	0.9994	–
Thermal	YOLO	0.9197	0.8509	0.9446	0.8590
	ThermalYOLO	**0.9945**	**0.8851**	**0.9958**	**0.8696**

Fig. 10 shows the BB estimation on the test dataset. The images are taken from different perspectives at the same time. The ground truth prediction made by YOLO on VSIs is displayed in red. The BB extracted from ThermalYOLo is shown in green.

As mentioned before, the people detection of YOLO on raw TIs is far from being similar to the performance on VSIs. Table 3 shows the percentage of images and the IoU of YOLO and ThermalYOLO on visible and TIs from the training, validation, and test datasets.

On the one hand, on VSIs, only YOLO was evaluated. It achieves SOTA performance with a 99.45% of obtained detection. This is taken as the ground truth.

On the other hand, YOLO and ThermalYOLO were compared in TIs. In this case, ThermalYOLO outperforms YOLO in terms of percentage by 8% in training and validation images, and 5% in test images. In both cases, values are very close to 100%. When comparing the IoU value, ThermalYOLO also outperforms YOLO, but the difference is lesser: around 3% in training and validation images and 1% in test images.

From these results, it can be stated that ThermalYOLO is necessary when dealing with TIs. It allows recognizing the person in the scene in almost 100% of the images. This is a critical step in some HPE top-down approaches, where first the person is detected, and only when this detection is successful, the key points are inferred. Otherwise, the second part of the algorithm (extracting the key points) is not executed. If a high rate of people detection is not critical in the system, YOLO could be used even in TIs. The IoU value is very similar to ThermalYOLO, but the retraining process would be avoided.

5.2. Obtaining 2D pose

Similar to the BB estimation, visible spectrum 2D HPE frameworks do not perform well on TIs, so it was necessary to develop a 2D HPE solution. In [27], authors proposed a method to obtain 2D poses on TIs directly. The core of the method is a pre-processing step that removes the background and centres the person in the image. Using Blazepose, the ground truth 2D poses are extracted directly from the TIs. Then, a NN regressor is built, following the same idea as described in Section 3.3.3. In the current work, a total of four backbone architectures were tested: MobileNetV2, ResNetV50, InceptionResNetV2 and VGG19. MobileNetV2 and ResNetV50 were also studied in [27]. In Table 4, both methodologies are compared. It shows the performance of different backbone architectures. Metrics regarding the NN complexity (number of parameters) and training details (time per epoch, total number of epochs to train) are displayed. In addition, the RMSE error in validation and test datasets. Also, fine-tuning and from-scratch configurations were tested, so the best configuration in each backbone is specified in the columns pre-trained and freeze. Finally, the validation and test errors are also exposed.

The configuration achieving the least validation and test errors in the current approach is the InceptionResNetV2 (0.0368 and 0.0337). However, it is also the most time-consuming architecture to train (45.77 s per epoch) since it has the highest number of parameters (5,201,338 or 5201M). ResNetV50 and VGG19 architectures have similar parameters (between 20M and 25M), but their performance is slightly worse than before. All these configurations have been trained from scratch. Unlike these, MobileNetV2 (the most lightweight architecture, having 5% of InceptionResNetV2's parameters) follows a transfer-learning approach and performs similarly to ResNetV50 and VGG19 in the validation dataset. However, its performance drops in the test dataset. Most complex configurations can better generalize on a dataset with different conditions. In addition, they find it most appropriate to train from scratch, learning the patterns and features of the TIs more helpful to perform the 2D HPE. Their high quantity of parameters can learn all of them and be fully adapted to TS particularities. MobileNetV2, as it has fewer parameters, follows a transfer-learning approach. Despite this, its performance is not much worse than the other architectures.

Using the methodology defined in [27], results show a similar pattern. The best-performing architecture is InceptionResNetV2 (0.0870 RMSE value on the test dataset). The rest of the architectures obtain similar values, ranging from 0.0911 to 0.0925. Compared to the current work, the results are much worse, mainly due to the smaller number of training images.

In this proposal, the 2D HPE is an intermediary step. The MobileNetV2 backbone would be appropriate if the training/inference time is critical, as it is a lightweight architecture that can be executed in real-time using mobile devices. In terms of 2D HPE performance, it is slightly worse than the others, but the 2D key points are only used to help the 3D HPE NN to infer the 3D pose, so having a non-optimum 2D HPE accuracy is not crucial.

In our case, as the solution has not been deployed on mobile devices and the training time is not a problem, InceptionResNetV2 was considered to obtain the 2D key points.

Fig. 11 shows the 2D HPE on test images. The images show the BB in blue colour. The ground truth 2D pose is displayed in red, while the green lines represent the 2D HPE using the proposed approach. The images show that the 2D HPE using InceptionResNetV2 is visually very close to the ground truth. It is important to remark that only one 2D HPE NN has been trained, handling different person perspectives.

Table 4
Performance of NN configurations in 2D HPE.

NN	NP	Approach	TE	NE	TL	RMSE Val	RMSE val
MobileNetV2	2,991,514	Current	7.22	106	Yes	0.0461	0.0452
		[27]	0.57	259	Yes	0.0757	0.0915
ResNetV50	24,691,546	Current	31.82	20	No	0.0464	0.0377
		[27]	2.43	146	No	0.07256	0.0925
InceptionResNetV2	**55,201,338**	**Current**	**45.77**	**134**	**No**	**0.0368**	**0.0337**
		[27]	1.42	392	Yes	0.0720	0.0870
VGG19	20,364,698	Current	43.55	22	No	0.0459	0.0384
		[27]	1.40	243	Yes	0.0747	0.0911

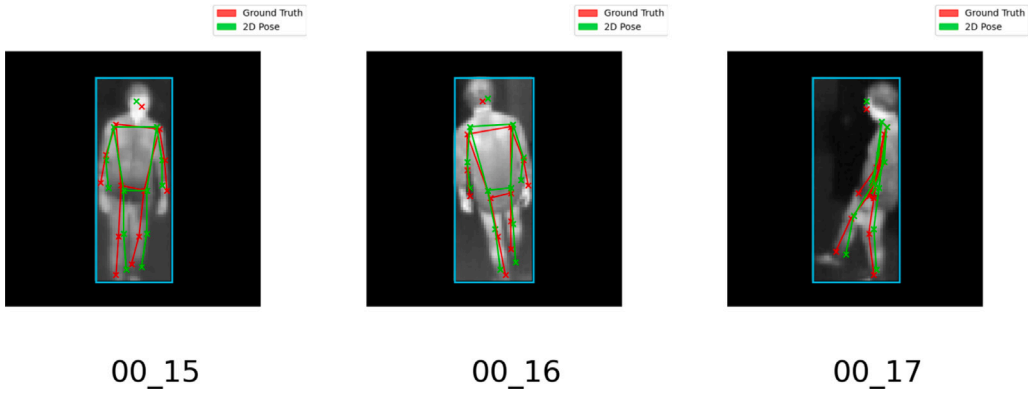

Fig. 11. 2D pose estimator's performance on test images.

Table 5
3D HPE solutions on VSIs.

Cameras	Percentage pose detected	
	VoxelPose	BlazePose
15	69.90%	**99%**
16	59.22%	**100%**
17	72.23%	**93.20%**
16-17	81.36%	–
15-16-17	84.27%	–

5.3. Obtaining 3D pose

In this work, the main goal is to obtain the 3D HPE on TIs. Since there are no existing solutions in the literature that perform this, existing 3D HPE solutions are executed on VSIs to obtain the ground truth of the paired TIs. Although many 3D HPE solutions exist, most are not publicly available, and only a few provide pre-trained models ready to be used with new datasets. For this reason, we could only execute two of them: VoxelPose and BlazePose. The first was available in the MMPose library, and the second in the Mediapipe library.

Table 5 shows the percentage of poses obtained by both solutions in the visible spectrum. On the one hand, VoxelPose achieves a maximum rate of 72.23% when following a single-view approach. On the other hand, BlazePose reaches 100% of poses in camera 00_16 and is always higher than 93%. Therefore, BlazePose outperforms VoxelPose when using only one camera. Nevertheless, VoxelPose can also find the 3D pose following a multi-view approach (more than one camera). When using two cameras, the percentage is 81.36% and using all the available cameras, 84.27%. A pose recognition of 84.27% would allow our system to establish VoxelPose as the ground truth 3D HPE method. However, the fidelity of the generated poses according to reality was far from accurate.

Fig. 12 compares the 3D ground truth poses obtained by VoxelPose and BlazePose in the same VSIs. It can be appreciated that VoxelPose can generate a pose, but it is not very well related to the actual pose of the user. However, BlazePose shows a better 3D representation of it. This is a recurrent pattern in all the images of the dataset. Furthermore, VoxelPose sometimes detects more than one person when only one person is in the scene. BlazePose was finally selected to obtain our proposed system's ground truth 3D poses. It is important to remark that in the future, if there is available some other SOTA 3D HPE techniques, they could be incorporated into this system.

Once the ground truth 3D poses were obtained in the VSIs, the configurations and NN architectures described in Section 3 were evaluated. Results are displayed in Table 6. From each configuration and NN backbone, the column 'Pre-trained' indicates if the convolutional layers had been pre-trained before on ImageNet, and a fine-tuning process freezing them was carried out to evaluate their performance. Furthermore, the validation and test RMSE errors are displayed.

Overall, only the 3-channel configurations support the pre-training of the convolutional part. This is because the used library (Keras) only allows transferring weights from 3-channel images when dimensions are not 224×224. The MobileNetV2 architecture is the only architecture that shows better behaviour when it is pre-trained. The rest of the backbones are much better when trained from scratch. The reason of this is the same as the case of the 2D HPE.

In the case of the 3-channel configuration architectures, the InceptionResNet achieves the best results (0.0513 RMSE value in the test dataset), closely followed by the ResNetV50 architecture (0.540). MobileNetV2 and transformer architectures are far from them (0.0698 and 0.0761, respectively). Architectures with the inception and residual approaches are suitable for learning the patterns and features of TIs that help infer the 3D pose. We note that transformer model weights are not pre-trained, and learning 3D pose from scratch is a challenging problem with a lower but similar performance to the other trained models. On the other hand, pre-trained Depth Wise Convolutions of MobileNetV2 (on the spectrum visible) are not straightforwardly related to 3-channel, which fuse multi-view thermal frames.

In Section 3.3.3, the 2D pose estimation is performed with the aim of helping the 3D pose detector to identify the 3D pose of the user in the scene. The impact of using the 2D pose is also evaluated in Table 6. When these architectures incorporate the 2D pose after the convolutional part, a noticeable improvement is observed in the

Fig. 12. Comparison between VoxelPose and BlazePose 3D HPE.

MobileNetV2 and transformer architectures (from 0.0698 to 0.0572 in the test dataset in MobileNetV2 and 0.0761 to 0.0569 in the case of the transformer). However, their performance is still worse than ResNetV50 and InceptionResNetV2. ResNetV50 improves the results in the validation dataset, but the performance in the test dataset is the same. Nevertheless, InceptionResNetV2 has slightly worse performance in both cases. The reasonable of this is that models with suitable patterns for thermal images (in 3-channel) overcome the performance of 2D estimation, which does not offer support in learning. On the other hand, Mobile or Transformers learning is guided by 2D pose to improve 3D posture.

Regarding the configurations using three single-channel input images, their performance is slightly better than the three-channel configuration discussed before in the case of InceptionResNetV2 and ResNet50. This is caused because the input comprises single-channel images, each processed by convolutional blocks independently. This allows extracting features and patterns more adapted to each particular camera independently because each has different perspectives and intrinsic and extrinsic parameters.

In single-channel images, when utilizing MobileNetV2 as the core model (as opposed to InceptionResNetV2), the combination of 2D pose information with TIs is crucial. However, it is noteworthy that in this particular scenario, ResNet50 achieves the highest level of performance with an error rate of 0.0496 on the test dataset. Nevertheless, when it comes to the usage of 2D pose information, the distinction between using it and not using it is quite minimal, amounting to only 0.0004. Consequently, it can be concluded that incorporating 2D pose information does not yield statistically significant or valuable improvements in the context of 3D HPE.

Finally, the configuration incorporating only 2D poses as input of NNs performs poorly compared to the configurations discussed above. To obtain the 3D pose, the NN extracts the most relevant information from the input images using the convolutional layers. Using the 2D poses discards useful features such as fingers, head, and other practical details to infer the 3D pose. 2D poses are helpful when fused with the features extracted by convolutional layers, not alone.

In an additional discussion on the learning performance, we include the number of parameters, number of epochs (with early stopping of patience 50), and time per epoch for each model in Table 7. We highlight the lower complexity of MobileNetV2 and learning time in 3-channel + 2D, which is relevant to edge/fog approaches integration. Transformer offers a weight configuration and learning time to learn features of a complex design with attentional layers from scratch. Only 2D-landmarks approaches are very light in neural complexity and learning time, at the cost of less demanding results. In addition, we note the increment in learning epochs when integrating 2D landmarks together with multiple input channels, due to the force of the adaptation of spatial patterns in the NN models to suit 2D estimation.

Fig. 13 shows the 3D HPE of three instances from the test dataset. In each row, the three first figures display the person with the BB and the 2D HPE from the three cameras of the setup. The fourth figure in each row displays the 3D HPE. In all cases, the red pose is the ground truth and the green colour represents the 2D and 3D pose estimations.

The first row (case 1) shows when the user faces the camera with *id* 00_15. In all cases, the ground truth 3D pose was obtained using a single-camera approach by this camera (00_15). Therefore, the 3D pose would be very similar to this, but also considering the depth coordinate. In this case, the user is in front of it; all the body parts are visible and can be detected by the camera. Therefore, the 3D HPE is very accurate.

In the second row (case 2), the user is situated in front of the camera 00_15, but turning back. This is more challenging than the case before because there exist some occlusions that cannot be detected only with camera 00_15. Despite this, the performance is very similar to the previous case.

Table 6
Performance of NN configurations in 3D HPE.

	NN	TL	RMSE val	RMSE test
3-channel	MobileNetV2	No	0.1005	0.0990
		Yes	**0.0549**	**0.0698**
	InceptionResNetV2	**No**	**0.0392**	**0.0513**
		Yes	0.0619	0.07487
	ResNet50	**No**	**0.0395**	**0.0540**
		Yes	0.0566	0.0760
	Transformer	No	0.0560	0.0761
3-channel + 2D	MobileNetV2	**No**	**0.5312**	**0.0572**
		Yes	0.0489	0.0636
	InceptionResNetV2	**No**	**0.0402**	**0.0516**
		Yes	0.0526	0.0647
	ResNet50	**No**	**0.0388**	**0.0540**
		Yes	0.0495	0.0652
	Transformer	No	0.0427	0.0569
3x 1-channel	MobileNetV2	No	0.1005	0.0991
	InceptionResNetV2	**No**	**0.0445**	**0.0500**
	ResNet50	No	0.0364	0.0503
3x 1-channel + 2D	MobileNetV2	No	0.0534	0.0582
	InceptionResNetV2	No	0.0539	0.0573
	ResNet50	**No**	**0.0371**	**0.0496**
2D	**1CNN**	**No**	**0.072**	**0.0770**
	Resnet	No	0.074	0.0786

Table 7
Details of NN training in 3D HPE.

	NN architecture	NP	NE	TE (s)
3-channel	**MobileNetV2**	2,990,311	97	**2.23**
	InceptionResNetV2	55,200,135	210	14.66
	ResNet50	24,690,343	**92**	9.96
	Transformer	14,937,959	161	42.09
3-channel + 2D	MobileNetV2	3,030,247	440	**9.34**
	InceptionResNetV2	55,240,071	188	14.42
	ResNet50	24,730,279	**146**	10.01
	Transformer	14,941,001	602	42.13
3x 1-channel	**MobileNetV2**	4,300,455	**17**	26.01
	InceptionResNetV2	56,772,423	240	**14.31**
	ResNet50	26,781,223	401	27.59
3x 1-channel + 2D	MobileNetV2	4,340,391	354	**25.96**
	InceptionResNetV2	56,812,359	482	40.59
	ResNet50	26,821,159	**226**	27.59
2D	**1CNN**	8,934,567	**74**	**0.74**
	Resnet	2,564,303	204	1.27

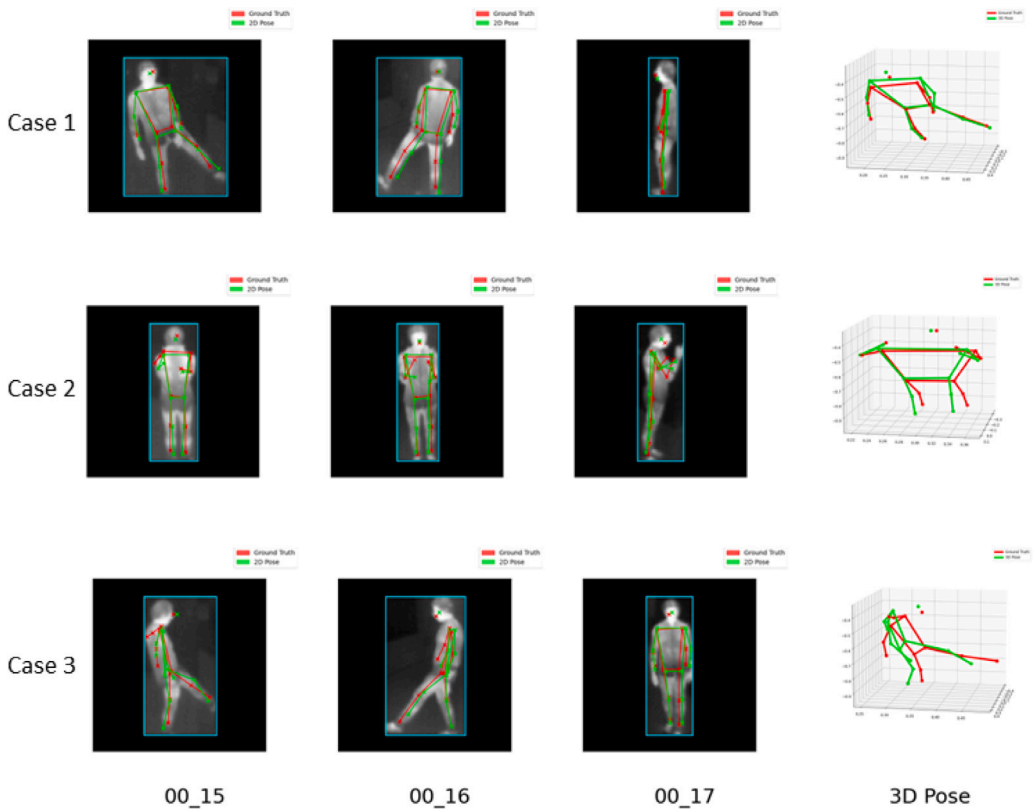

Fig. 13. 3D pose estimator's performance on test images.

In the last row (case 3), the user is in profile with the camera 00_15. This is the most challenging case because there are occluded parts that also have different depths. In addition, half of the user can be occluded, being hard to identify the pose of the body part that is not situated in front of the camera. In spite of this, the proposed 3D approach is able to learn the relation between cameras to infer the depth of the different body parts and show accordingly.

Finally, among the 3D HPE methods, multi-view methods have been demonstrated to produce better performance than single-view, with the drawback of requiring more complex scenarios and more costly models. In this work, the last experiment involves the comparison between single-view and multi-view models setting as baseline the architectures defined in Section 3.3.4. Table 8 shows the RMSE value of the models incorporating images from the different permutations of cameras.

The single-view approach produces values between 0.0571 and 0.0529. These are higher than those using more than one camera. Thus, it is also demonstrated in this work that having different perspectives of the same person in the scene helps the model to better locate the 3D key points. In the multi-view approach, the configuration using 3 cameras (0.0496 of RMSE value) outperforms the solutions using only two cameras. Regarding the best architecture in each case, InceptionResNetV2 is the most complex architecture and produces the best results in two single-view and two-view configurations. In the single-view approach using camera 15 and the configuration using three cameras, the best-performing architecture is ResNetV50.

6. Conclusions

This work faces the problem of 3D HPE in TIs. Current SOTA solutions in the visible spectrum are not well adapted to thermal images and no robust specifically designed solutions to the thermal spectrum have been proposed in the literature. In our methodology, we propose the use of several cameras following a multi-view approach, increasing the ability of the solution to infer 3D poses and avoiding occlusions.

An IoT device incorporating 2 cameras is proposed to record images. Images from several devices are synchronized using the MQTT protocol and the homography matrix is used to pair the visible and thermal images in each device. The dataset is publicly available to provide a benchmark for similar works TIs.

The bounding box is obtained in the TIs after fine-tuning YOLOv3. Although the training time was lesser, using the pre-trained weights on the COCO dataset did not achieve good results. Training from scratch allows learning the thermal-specific features, leading to lesser error. Compared to the original YOLOv3, ThermalYOLO achieves more than 5% bounding box recognition and 1% of IoU value in the test dataset.

Once obtained the BB, the TI is pre-processed (cropping and centring the person). The 2D key points are obtained using a 2D pose estimator trained to obtain the ground truth 2D poses obtained by BlazePose. The 2D pose NN incorporates the InceptionResNetV2 backbone as it has the least RMSE value (0.0337) compared to other convolutional architectures.

Finally, the 3D pose making use of three cameras is obtained. Several configurations have been thoroughly compared. Among them, processing the TIs independently using ResNet50 backbones, fusing the output, and concatenating the 2D poses achieved the best results (0.0496 RMSE value in the test dataset). Visual comparisons demonstrate that the 3D pose obtained following the multi-view approach is able to outperform single-view approaches in high occlusion cases.

In future works, the proposed NNs will be deployed in IoT devices to enable a real-time 3D HPE. To enable this, some big data frameworks such as Apache Spark, and Apache Kafka will be deployed. In addition, accelerators such as Coral.AI will be embedded to the IoT devices to speed up their inference time. Furthermore, the gathering of ground truth 3D poses will be improved. Instead of using some pre-trained single-view solutions, some tags will be attached to the users' joints to obtain accurate 3D poses.

Table 8
Comparison between single and multi-camera approach.

Approach	N. cameras	Camera	Architecture	RMSE test
Single-View	1	15	ResNet50	0.0571
		16	InceptionResNetV2 + 2D Pose	0.0546
		17	InceptionResNetV2	0.0529
Multi-View	2	15-16	InceptionResNetV2	0.0509
		16-17	InceptionResNetV2	0.0513
		15-17	InceptionResNetV2	0.0498
	3	**15-16-17**	**ResNet50 + 2D Pose**	**0.0496**

CRediT authorship contribution statement

Marcos Lupión: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Aurora Polo-Rodríguez:** Methodology, Software, Investigation, Writing – review & editing. **Javier Medina-Quero:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – review & editing, Supervision, Funding acquisition. **Juan F. Sanjuan:** Methodology, Validation, Resources, Writing – original draft, Supervision. **Pilar M. Ortigosa:** Conceptualization, Methodology, Validation, Resources, Investigation, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data is available in Mendeley Data.

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